Detecting Cafeteria Waste with Specialized YOLO11 Models

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Abstract

Automated cafeteria waste sorting involves two distinct tasks: identifying discarded objects and determining the disposal stream appropriate for each one. This study evaluates a prototype that combines four specialized YOLO11 detectors, covering seven cafeteria-item classes, with a rule-based policy that suggests composting, recycling, or manual review. The frozen models were evaluated on 186 validation photographs and a historically exposed cohort of 248 photographs containing 924 AI-assisted object annotations. On the latter cohort, the system achieved 87.78% mAP50 and 67.24% mAP50–95. At a uniform confidence threshold of 0.50, micro precision was 93.42% and recall was 78.35%. Selecting separate thresholds for each class on validation increased precision to 97.03% on validation and 97.22% on the historical cohort, while recall declined to 71.43% and 68.18%, respectively. Historical-cohort F1 fell from 85.23% to 80.15%. Only six historical images originated from the participating school, and the available records contain no independently verified disposal-bin annotations. The results therefore demonstrate how confidence selection changes recognition performance; they do not establish correct-bin accuracy or field generalization.

Keywords: object detection, YOLO11, cafeteria waste, selective prediction, recycling.

1 Introduction

At the end of a school lunch period, discarded food, drink cartons, utensils, and packaging may be mixed together on a tray or in a waste container. Computer vision can help identify these objects, but identifying an item and deciding where to discard it are not equivalent problems. A plastic bottle might be recognizable from its shape while its eligibility for recycling still depends on its contents, contamination, and local collection rules. The U.S. Environmental Protection Agency recommends checking local recycling guidance and emptying and cleaning recyclable containers [1]. A system that detects a bottle correctly may therefore still recommend the wrong bin.

Object detection is a useful starting point because it locates individual items in scenes containing several kinds of waste. Its predictions, however, involve uncertainty. A low-confidence prediction may be incorrect, while rejecting it may leave a real item unidentified. This distinction becomes important when detection results are used to support disposal decisions: a system can increase the proportion of correct predictions simply by withholding more of them, even as the number of correctly identified objects decreases.

This study examines a prototype consisting of four specialized YOLO11 detectors and a separate rule-based disposal policy. It addresses three related questions: how well the retained detectors recognize seven cafeteria-item categories on available development images; whether selecting confidence thresholds by class improves precision without retraining; and what additional evidence is required before object detections can be interpreted as reliable waste-sorting decisions. The analysis uses frozen checkpoints and recorded predictions, allowing changes in decision thresholds to be evaluated independently of changes in the detectors themselves. Because the nominal test photographs have prior development exposure, the findings are presented as development results rather than prospective field performance.

1.1 Related Work

Early approaches to waste recognition often classified the contents of an entire photograph. Thung and Yang's TrashNet project focused on individual waste items against comparatively simple backgrounds [2], while Single, Iranmanesh, and Raad introduced RealWaste using material photographed at a landfill [3]. These datasets illustrate how differences in image collection and object appearance affect the task being studied. Neither whole-image classification setting fully represents a cafeteria scene containing multiple overlapping items.

Localization datasets address a different aspect of waste recognition. Proença and Simões introduced TACO, which annotates litter instances within their surrounding environment [4]. Bashkirova and colleagues developed ZeroWaste to examine deformable and overlapping materials in industrial recycling imagery [5]. These studies are relevant to the crowded scenes encountered in waste collection, although recognizing an object or material remains different from verifying whether a particular school accepts it for recycling or composting.

The detector family used in this project originates from the YOLO framework introduced by Redmon and colleagues [6]. We use the YOLO11 implementation released by Ultralytics [8] and measure localization performance using the average-precision conventions associated with the COCO benchmark [7]. Average precision evaluates confidence-ranked detections over multiple operating points; it does not state how many correct and incorrect detections a system will retain after choosing a particular cutoff. For that reason, fixed-threshold precision and recall are also central to this study.

Selective classification provides a related framework for deciding when to withhold a prediction. Geifman and El-Yaniv examined confidence-based rejection [9] and subsequently introduced SelectiveNet, which jointly trains

prediction and rejection components [10]. In contrast, the present work does not train a rejection network. It applies different confidence thresholds to existing detectors and measures the resulting precision–recall tradeoff. Finally, research on distribution shift, including the WILDS benchmark [11], and on data leakage in scientific machine learning [12] highlights the importance of separating development decisions from independent evaluation. These concerns motivate our explicit audit of the historical image and annotation records.

2 Data

The working manifest contains 2,372 training views, 186 validation photographs, and 248 photographs in a historical test partition. The training count includes 2,052 original image entries and 320 additional augmented views. Twenty original photographs submitted by the participating school contribute 340 training views when their recorded augmentations are included. Augmented views present modified versions of existing photographs; they are not independent observations. Moreover, the four specialist checkpoints have different training histories, so the current manifest should not be described as a single common dataset used to train all four models from scratch.

The images were assembled from several public datasets and school submissions. The historical 248-image partition contains 73 images from a dataset labeled school_lunch, 66 from tetra_pak, 48 from trash_detection, 31 from plate, 20 from utensil, four from macworx, and six from the participating school's own collection. Thus, only 6/248 images (2.42%) were collected at that school; the separately named school_lunch source does not belong to the school's submission. The historical sample is consequently dominated by public-source imagery rather than representative observations of the intended cafeteria environment.

Table 1 summarizes the class distribution. The 248-image partition contains 924 annotated objects, but the distribution is highly uneven: food accounts for 433 objects, whereas hinged containers account for only four. The validation partition likewise contains just two hinged-container annotations. Strong average precision for such a sparsely represented class should therefore be interpreted cautiously.

The evaluation uses the retained, repaired annotation manifests rather than superseded label copies. A previous audit documented discrepancies in 109 older validation-label files, including ten files with bottle-label differences. Annotations were produced or reviewed with AI assistance, and comprehensive independent human verification is not documented. Ambiguous serving boundaries, visually similar packaging, and incomplete labels may consequently affect the measured results. No labels were modified for the present manuscript analysis.

Table 1: Object boxes by class (training views include augmentation; Hinge means hinged container).

Class	Train	Val.	Test
Food	2,501	343	433
Plate	722	110	123
Utensil	457	101	98
Hinge*	185	2	4
Carton	972	94	134
Wrapper	316	85	75
Bottle	293	42	57

3 Methods

The prototype combines four YOLO11 specialists, with each checkpoint responsible for a specified subset of the seven target classes (Table 2). One model detects food, hinged containers, and drink cartons; another detects plates and wrappers; the remaining two detect utensils and bottles separately. These divisions reflect the previously developed system. The present experiment does not establish that four specialists outperform a unified seven-class model, since no controlled comparison of those architectures was performed. Checkpoint hashes, class mappings, and inference settings were recorded in the research freeze at commit defcc6156a4425daf1ae530757634cc541f3c267.

Table 2: Frozen specialist models. Input size is given in pixels; Hinge denotes hinged container.

Model (input)	Parameters	Classes
YOLO11n (640)	2,590,425	Food, hinge, carton
YOLO11n (640)	2,605,126	Plate, wrapper
YOLO11x (800)	56,877,241	Utensil
YOLO11s (640)	9,428,179	Bottle

Inference was performed with Ultralytics 8.4.175, PyTorch 2.14.1, and Python 3.12. Each checkpoint processed the same image independently at its recorded resolution. Test-time augmentation was disabled; each specialist applied non-maximum suppression with an IoU cutoff of 0.70 and a limit of 300 detections. Outputs were then mapped to the seven shared class names and concatenated. Because no additional suppression was performed across specialists, overlapping predictions from different categories could remain in the combined output. This detail matters when one physical item resembles more than one target class.

The prototype includes a separate rule-based disposal layer rather than a learned bin classifier. At the default display threshold of 0.25, detections below confidence 0.50 are referred for review. Above 0.50, food detections receive a compost suggestion, while bottle and drink-carton detections receive a recycling suggestion. Plates, utensils, wrappers, and hinged containers are always marked for review. No category produces an automatic landfill suggestion. These rules are experimental: the system does not verify an item's material, whether a carton is empty, or the school's specific collec-

tion requirements. A suggestion therefore describes the rule applied to a recognized object, not independently established disposal eligibility.

The recorded evaluation environment was an Apple M1 MacBook Air with 8 GiB of memory and MPS acceleration available. Neither inference latency nor energy use was measured. The retained checkpoint and configuration hashes were checked during the research audit so that the results below refer to a fixed system rather than a new training run.

4 Evaluation

Object detection was measured using intersection-over-union (IoU), defined as the area shared by a predicted bounding box and its annotation divided by the area of their union. We report mAP50, which averages precision across classes at IoU 0.50, and mAP50–95, which additionally averages across IoU thresholds from 0.50 through 0.95 [7]. These ranking-based metrics are distinct from the precision of predictions retained at one chosen confidence threshold.

For the fixed-threshold comparisons, predictions were considered in descending confidence order and matched one-to-one to unused annotations of the same class at IoU at least 0.50. Each successful match counted as a true positive (TP); unmatched predictions and unmatched annotations counted as false positives (FP) and false negatives (FN), respectively. Micro precision, recall, and F1 pooled those counts across all seven classes:

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2PR}{P + R}.$$

An additional evaluation concerns the disposal-suggestion layer. For food, bottle, and drink-carton detections, classwise matching counts whether an automatic suggestion corresponds to an annotation of the same object class. A more conservative geometry-first match instead assigns predictions to annotated boxes before checking class agreement, exposing competition between overlapping specialists. Both methods measure object-recognition proxies: neither can measure correct-bin sorting without verified disposal labels. We also distinguish precision among accepted predictions from coverage of all annotated objects, because missed detections generate neither suggestions nor review requests.

The 186-image validation partition was used to choose operating thresholds. The 248-image historical partition was scored afterward without retuning those thresholds, but its photographs had already influenced earlier model development. Accordingly, these observations are not an independent test of generalization [12]. Confidence filtering changes the set of accepted detections and thus fixed-threshold precision and recall; it does not improve the checkpoint weights or the underlying ranking-based mAP.

5 Threshold Selection

We first evaluated whether a single confidence threshold could achieve high precision while retaining a useful number of detected objects. Global thresholds from 0.50 to 0.99, in increments of 0.01, together with 0.995, were evaluated on the 186-image validation partition. The selection criteria required at least 97% micro precision, 60% recall, and 100 accepted detections. No uniform cutoff met all three. Among the global thresholds satisfying the recall and detection-count requirements, the highest precision was 96.04% at confidence 0.79, with recall of 68.73%.

We next allowed the cutoff to vary by class. For each of the seven categories, candidate thresholds were summarized by the true and false detections they retained on validation. A dynamic program combined these possibilities, choosing the threshold vector that maximized accepted true positives while satisfying the same aggregate constraints:

$$\begin{aligned} \max_{\tau} \quad & TP(\tau) \\ \text{subject to} \quad & P(\tau) \geq 0.97, \\ & R(\tau) \geq 0.60, \\ & N_{\text{pred}} \geq 100. \end{aligned}$$

The resulting values appear in Table 3. Bottle received the strictest cutoff, 0.91, while utensil and hinged-container thresholds remained at 0.50. This method optimizes the mix of accepted detections on the validation set; it does not calibrate confidence scores as probabilities of correctness or guarantee that those thresholds will transfer to other image sources.

Table 3: Validation-selected confidence cutoffs by class; Hinge denotes hinged container.

Class	Cutoff	Class	Cutoff
Food	0.79	Plate	0.83
Utensil	0.50	Hinge*	0.50
Carton	0.84	Wrapper	0.83
Bottle	0.91	—	—

The selected vector was subsequently applied without modification to the historical cohort. It was not used to replace the existing disposal-policy settings, and no additional cutoff selection was performed on those historical images. Because those images had already informed development, the resulting comparison remains descriptive rather than an independent estimate of field performance.

6 Results

On the historical development cohort, the frozen system achieved 87.78% mAP50 and 67.24% mAP50–95, compared with 86.36% and 65.25% on validation. Results varied considerably by category (Tables 4 and 5). For example, drink cartons achieved 99.23% precision and 96.27% recall at confidence 0.50, whereas bottle recall was 57.89%. Hinged containers achieved 99.50% AP50,

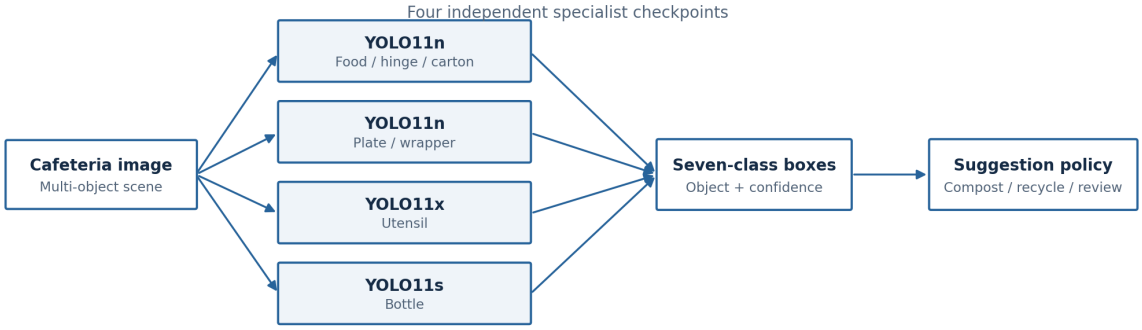


Figure 1: Processing pipeline: four YOLO11 specialists produce seven item classes, which are mapped to provisional compost, recycling, or manual-review suggestions.

Table 4: Detection precision, recall, and AP50 on 248 development images (%); Hinge denotes hinged container.

Class	Precision	Recall	AP50
Food	97.03%	75.52%	83.91%
Plate	87.29%	83.74%	87.04%
Utensil	90.91%	71.43%	86.06%
Hinge*	80.00%	100.00%	99.50%
Carton	99.23%	96.27%	97.55%
Wrapper	89.23%	77.33%	85.13%
Bottle	76.74%	57.89%	75.27%

Table 5: Detection counts, F1 and AP50–95 for the same cohort. TP/FP/FN are object counts; Hinge denotes hinged container.

Class	TP/FP/FN	F1	AP50–95
Food	327/10/106	84.94%	55.09%
Plate	103/15/20	85.48%	71.60%
Utensil	70/7/28	80.00%	61.26%
Hinge*	4/1/0	88.89%	91.17%
Carton	129/1/5	97.73%	74.28%
Wrapper	58/7/17	82.86%	57.64%
Bottle	33/10/24	66.00%	59.62%

but only four such objects were annotated in the historical cohort, limiting the interpretation of that figure. Across all seven classes, the historical cohort yielded 93.42% micro precision, 78.35% recall, and 85.23% F1 at confidence 0.50.

Table 6: Automatic-suggestion object metrics. Neither matching method establishes correct-bin accuracy.

Metric	Per class	Joint geometry
Matches / 510	489	483
Precision	95.88%	94.71%
Recall	78.37%	77.40%
F1	86.24%	85.19%

Class-specific matching counted 489 correct object detections among 510 automatic suggestions; geometry-first matching retained 483. Five food detections and one bottle detection were instead assigned to overlapping wrapper labels under the latter convention. These are disagreements in object assignment, not verified wrong-bin outcomes. The frozen 248-image predictions were reproduced in a separate inference replay, and the present analysis uses those saved outputs without retraining the models.

Table 7: Default sorting-policy outcomes. Denominators depend on detections versus labeled objects.

Outcome	Count (rate)
Compost	337/881 (38.25%)
Recycling	173/881 (19.64%)
All automatic	510/881 (57.89%)
Manual review	371/881 (42.11%)
Auto matched (IoU)	489/924 (52.92%)
Auto correct class	483/924 (52.27%)
Reviewed objects	286/924 (30.95%)
Missed objects	149/924 (16.13%)
Unmatched auto	21/510 (4.12%)
Unmatched review	85/371 (22.91%)

At the display threshold of 0.25, the default policy emitted 881 detections: 337 compost suggestions, 173 recycling suggestions, and 371 manual-review outcomes.

Automatic suggestions therefore accounted for 57.89% of emitted detections. This denominator is narrower than the set of actual objects in the photographs, since undetected items generate no suggestion or review record. Indeed, exact-class automatic matches account for 483 of the 924 annotated objects (52.27%), a coverage-sensitive detection proxy rather than correct-bin accuracy. We do not attach conventional object-independent confidence intervals to these results: predictions are clustered within photographs, annotations contain unresolved uncertainty, and the historical cohort has prior development exposure.

6.1 Precision–Recall Trade-off

Table 8: Seven-class detection at IoU 0.50. P, R and F1 are percentages.

Cohort / cutoff	TP/FP/FN	P	R	F1
Val. 0.50	638/48/139	93.00	82.11	87.22
Val. classwise	555/17/222	97.03	71.43	82.28
Hist. 0.50	724/51/200	93.42	78.35	85.23
Hist. classwise	630/18/294	97.22	68.18	80.15

Class-specific thresholds reduced false positives on the historical cohort from 51 to 18, but also reduced true positives from 724 to 630. Accepted predictions fell from 775 to 648. Precision increased by 3.80 percentage points, while recall fell by 10.17 points and F1 decreased from 85.23% to 80.15%. Validation F1 similarly declined from 87.22% to 82.28%. Thus, class-specific thresholding produced more precise accepted detections at the cost of missing more objects. It did not improve mAP or the underlying detectors.

7 Data Independence

The principal threat to independent evaluation is the possibility that photographs, closely related capture sessions, or earlier prediction results influenced model selection. We therefore examined the retained local inventory rather than inferring independence from folder names. The audit identified 102,806 readable image paths, 67,422 distinct filesystem inodes, and 14,379 byte-distinct images. Among 9,468 candidate identities, 3,842 were flagged for possible exposure, capture-group overlap, or duplication; 5,482 had unresolved historical provenance; and 144 original HEIC files could not be visually decoded in the audit environment. None could be positively certified as independent under the available records (Table 9), although this does not establish that every candidate was actually used to train a checkpoint.

To identify potentially repeated imagery, the audit compared file-level SHA-256 and decoded-pixel hashes, then screened near-duplicates using rotation- and reflection-aware comparisons of 64-bit difference hashes at a Hamming-distance threshold of five. The screening pool contained 4,751 images associated with training directories or prepared manifests. These checks provide conservative evidence of possible overlap, not exhaustive proof of model exposure: directory membership does not guarantee that a particular optimizer consumed an image, while nonmatching hashes cannot exclude related crops, viewpoints, or photographs taken in the same session.

Table 9: Provenance of 9,468 candidate image identities. Categories do not imply confirmed training exposure.

Candidate status	Images
Exposure/group/duplicate flag	3,842
Unresolved provenance	5,482
HEIC decoding unavailable	144
Confirmed independent	0

Both school submissions have development-related provenance. The earlier submission contained 123 photographs and the later one 21; each set shared recorded training or development capture groups. Four images from the October 29 submission entered the training collection, excluding that capture day from independent evaluation. Historical descriptions such as “sealed test” must therefore be interpreted alongside later inference receipts and model-use records. Overlap counts describe different audit indicators and cannot be summed as if they identified disjoint training exposures.

The audit separately examined annotation provenance, screening 2,398 JSON files together with additional CSV, XML, and YAML records. It did not identify a confirmed set of independently human-adjudicated, object-level disposal-bin labels. An annotation marked “reviewed” in an earlier workflow is not, by itself, evidence that a human established the object’s material or appropriate bin. Forty-two unreadable image aliases and five malformed or inaccessible JSON records were also documented rather than silently repaired. The complete provenance summary and file identities are retained in the research-readiness report and external audit artifacts.

Table 10: Independent sorting outcomes remain unmeasured; required evidence is shown.

Outcome	Required evidence
Bin precision/recall/F1	Independent images, human bin labels
Correct-bin accuracy	Verified local bin eligibility
Automatic object coverage	Complete independent annotations
Field review rate	Sealed school cohort

The absence of these metrics should not be interpreted as zero performance. They are undefined because the audited collection does not contain the independent disposal-label evidence needed to compute them.

8 Error Analysis

We examined the frozen predictions for food and bottle, two classes that contribute to automatic disposal suggestions. A missed object was assigned a diagnostic category according to the strongest same-class overlap in the saved outputs: a sufficiently localized detection

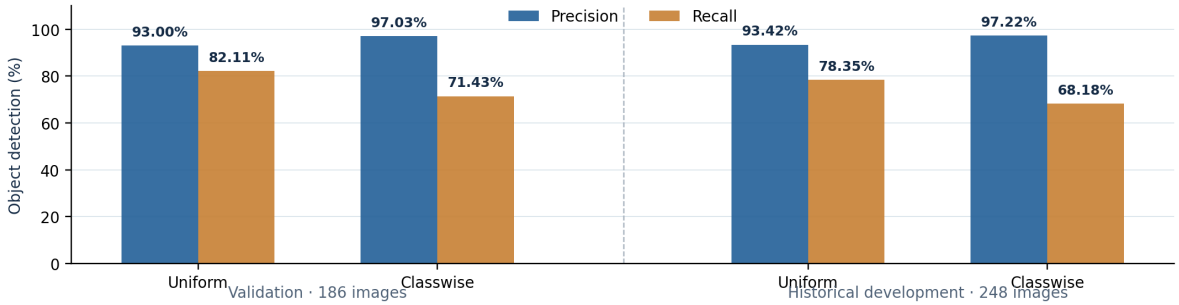


Figure 2: Precision and recall at uniform and class-specific thresholds, compared separately on validation and previously examined historical images.

below confidence 0.50, a candidate overlapping at IoU 0.10–0.49, or no qualifying same-class overlap. False positives were similarly grouped by localization, duplication, competing class, or lack of a corresponding annotation. These categories are algorithmic descriptions rather than verified causal explanations. In particular, an apparent background false positive could reflect an incomplete reference annotation.

Table 11: Food and bottle error diagnostics at confidence 0.50. Categories are provisional.

Error category	Food	Bottle
Correct matches	327	33
FN: below threshold	53	19
FN: localization	48	1
FN: no overlap	5	4
FP: localization	2	0
FP: duplicate	1	0
FP: background/label	7	9
FP: class ambiguity	0	1

Errors were concentrated in a few demanding photographs. One buffet scene accounted for 52 of 106 missed food objects (49.06%), while a single outdoor-trash scene accounted for 10 of 24 missed bottles (41.67%). The totals therefore reflect substantial within-image clustering rather than independent failures across dozens of different scenes. Both photographs were retained in the evaluation.

A targeted AI-assisted visual inspection considered 12 error cases from ten unique photographs. The examples included a sports-style bottle outside the intended disposable-container scope, a possible unannotated bottle, ambiguous cylindrical packaging, crowded food portions, and sealed food packaging recognized as food. These observations help identify plausible annotation and class-boundary problems, but cannot determine actual material, contents, or bin acceptance from the available previews. No annotations or disposal rules were modified.

A previously completed bottle-specialist trial also provided a negative experimental result. One additional epoch of classification-head training decreased validation AP50 from 82.47% to 77.95% and AP50–95 from 62.00% to 59.18%. The candidate failed its predefined validation criterion and was neither activated nor scored on the historical cohort. All main-paper results continue to use the frozen bottle checkpoint.

8.1 Additional Image Evaluation

The unchanged detectors were also applied to 44 source-screened public photographs: nine from bottle sources, six from carton sources, fifteen from utensil sources, twelve from hinged-container sources, and two from wrapper sources. At confidence 0.25, the system emitted 149 predictions—twelve compost suggestions, 49 recycling suggestions, and 88 review outcomes—while three photographs produced no displayed detection. These are output counts, not measurements against independently verified object or disposal labels.

A preliminary source-category check found at least one detection of the expected source class in 9/9 bottle photographs, 5/6 carton photographs, 12/15 utensil photographs, 4/12 hinged-container photographs, and 0/2 wrapper photographs. These proportions are not class recall: the source categories do not provide exhaustive instance-level ground truth, and several photographs contain multiple objects. Visual examination suggested repeated confusion between open foam containers and plates, missed wrappers, and overlapping predictions in crowded bottle scenes. In one example, a foil meal tray was classified as a drink carton with confidence 0.617 and consequently received a recycling suggestion; the photograph alone does not establish its correct local disposal stream.

A separate, provisional comparison used approximate AI-generated boxes for 41 of the 44 photographs, excluding three crowded bottle scenes. The frozen V1 model ensemble achieved 46.25% mAP50 and 21.16% mAP50–95, while a developmental hinged-container V3 candidate achieved 44.78% and 21.47%, respectively. Because the annotations were not independently human verified and the photographs had already been inspected during development, neither comparison constitutes a valid independent benchmark. The V3 candidate was not incorporated into the frozen system.

Taken together, these observations illustrate plausible failure modes and the limits of the available annotation

process, not demonstrated field accuracy or a validated improvement under distribution shift. Case-level notes are preserved in the archived visual audit. The source images require rights verification before reproduction in a publication.

9 Limitations

The first limitation is evaluation independence. Although the model configuration and reported outputs are frozen, the historical cohort was consulted during earlier development, and the full lineage of every candidate photograph cannot be reconstructed. AI-assisted object annotations introduce further uncertainty, particularly in scenes where serving boundaries or similar forms of packaging are ambiguous. These factors weaken claims of generalization even when a benchmark's numerical precision appears high. The broader literature on distribution shift and leakage underscores why reused or mismatched evaluation data may overstate performance in a new setting [11, 12].

A second limitation is the gap between object recognition and sorting correctness. The policy does not establish whether bottles and cartons are empty, whether packaging contaminates food scraps, or whether the local waste contractor accepts a particular material. Nor does the system automatically recommend landfill disposal. Manual review is an abstention, not a successful sort, while undetected objects are omitted from the displayed review rate. No evidence was collected about contamination reduction, user compliance, inference latency, or energy consumption in a live school deployment. Public-dataset licensing and attribution must also be verified before any image redistribution.

An independent field study would require photographs collected in previously unused school sessions, with inclusion criteria and capture-group boundaries fixed before inference. Human annotators blinded to model predictions should identify each eligible object and document its actual disposal destination according to school-specific collection rules and, where necessary, material or contents evidence beyond the photograph. Disagreements and uncertain items should undergo recorded adjudication. The frozen checkpoints, thresholds, and decision rules could then be evaluated without further tuning, with class-specific detection metrics, correct-bin precision and recall, manual-review workload, and automatic coverage reported together. In addition to accuracy among accepted recommendations, the study should report the proportion of all eligible objects correctly assigned without review; otherwise, abstention could make a weak system appear artificially accurate. Uncertainty should be estimated across independent capture groups rather than treating objects from one image as independent observations.

10 Conclusion

This study evaluated a four-specialist YOLO11 system for recognizing seven categories of cafeteria waste. On 248 previously examined development photographs, the system achieved 87.78% mAP50 and 67.24% mAP50–95. Class-specific confidence thresholds increased object-detection precision to 97.03% on validation and 97.22% on the historical cohort, but lowered recall and F1. These results demonstrate the trade-off between accepting fewer false detections and missing more objects.

A central limitation is that the available images do not form an independent school-based test set, and verified bin labels are unavailable. The next step is to collect new cafeteria photographs and evaluate the unchanged system against independently reviewed object and disposal annotations. Only then can its effectiveness as a waste-sorting assistant be established.

11 Reproducibility

The project repository retains the inference code, class mappings, disposal rules, evaluation scripts, and summary results. The recorded configuration in configs/research-freeze.json identifies the four checkpoint hashes and fixed inference settings; results/sorting-verification.json contains the retained detection results, and results/research-readiness.json summarizes the provenance audit. Evaluation conventions and label requirements are described in docs/sorting-evaluation.md.

The complete image data, checkpoint binaries, and detailed local audit artifacts are held outside Git in the project's accompanying asset directory. The working data manifest and frozen repaired annotation files are likewise external. As a result, the public source code alone does not reproduce native inference. Historical source and license declarations are retained with the assets, but attribution and redistribution permissions have not been independently audited; the school photographs are not presented as a released dataset.

The repository documents the commands for reproducing the file audit, running software tests, and evaluating newly collected annotations. These procedures do not retrain the frozen detectors or establish independent sorting accuracy; the latter requires a separately collected, human-verified evaluation cohort.

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13 Acknowledgments

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